

AI Energy Systems Report

Can Global Energy Systems Sustain the Rapid Growth of AI?

AI, Electricity Demand & Energy System Constraints

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Selected Country / Region: World

Selected Year: 2025

Executive Summary

This report presents an integrated analysis of global electricity demand, AI energy scenarios, readiness rankings, grid stress, and infrastructure requirements through 2035.

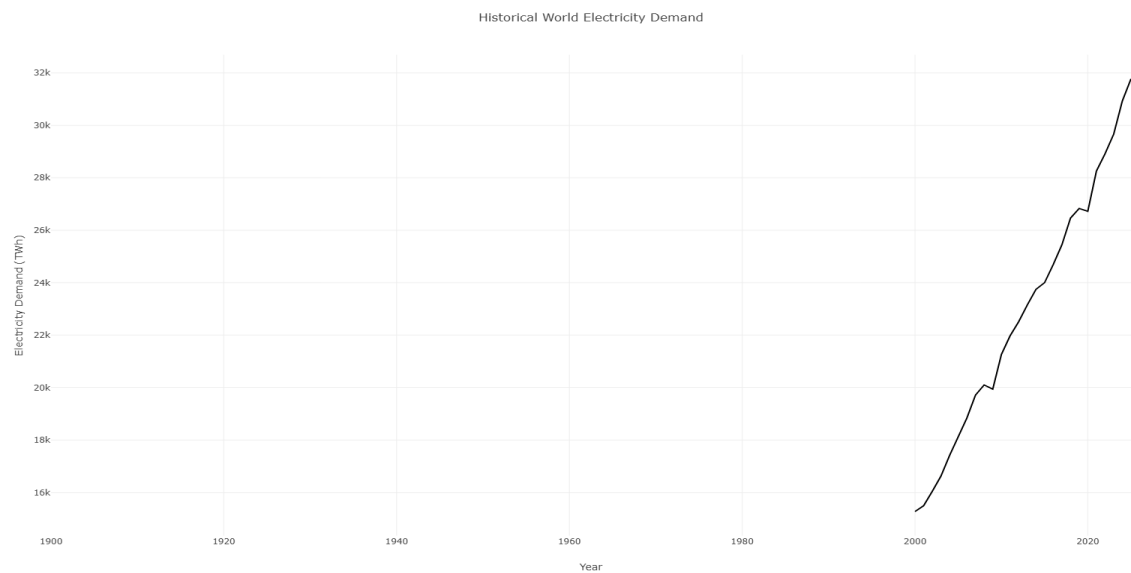
Latest World Electricity Demand: 31,772 TWh

Global Carbon Intensity: 458.29 gCO₂/kWh

Forecasted 2035 World Demand with AI Impacts: 39,655 TWh

Highest AI Demand Scenario (2035): 1,719 TWh

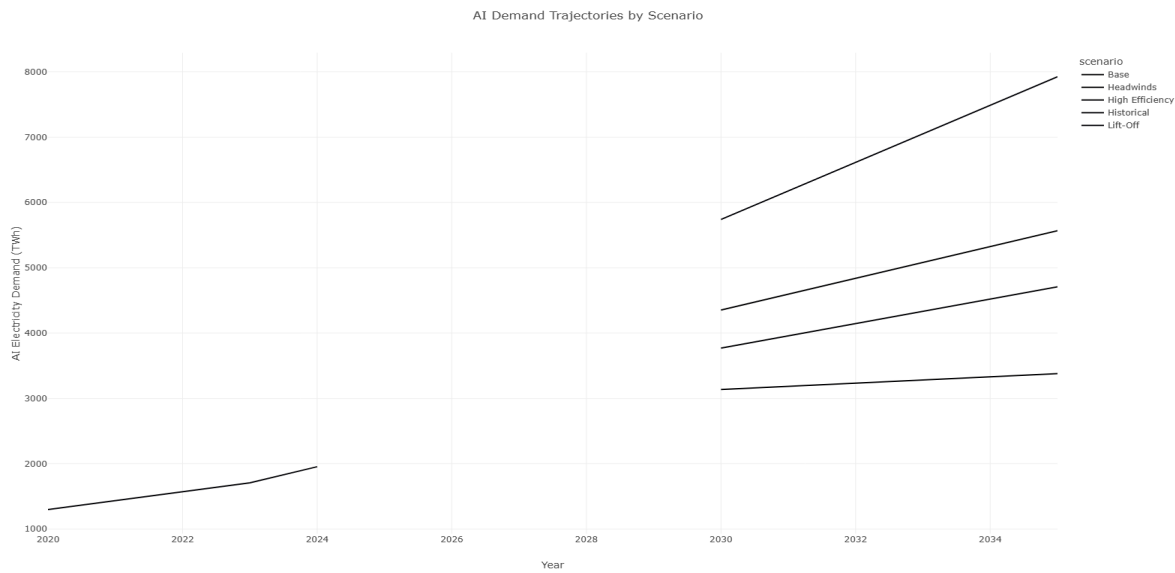
Global Electricity Demand & Regional Analysis



Top 10 Electricity Demand Regions in 2025

Country	Electricity Demand
World	31,772.35
G20 (Ember)	26,769.54
Asia (Ember)	16,728.88
OECD (Ember)	11,365.01
China	10,563.90
G7 (Ember)	7,829.55
North America (Ember)	5,172.82
Europe (Ember)	5,101.69
Europe	4,598.53
United States	4,535.81

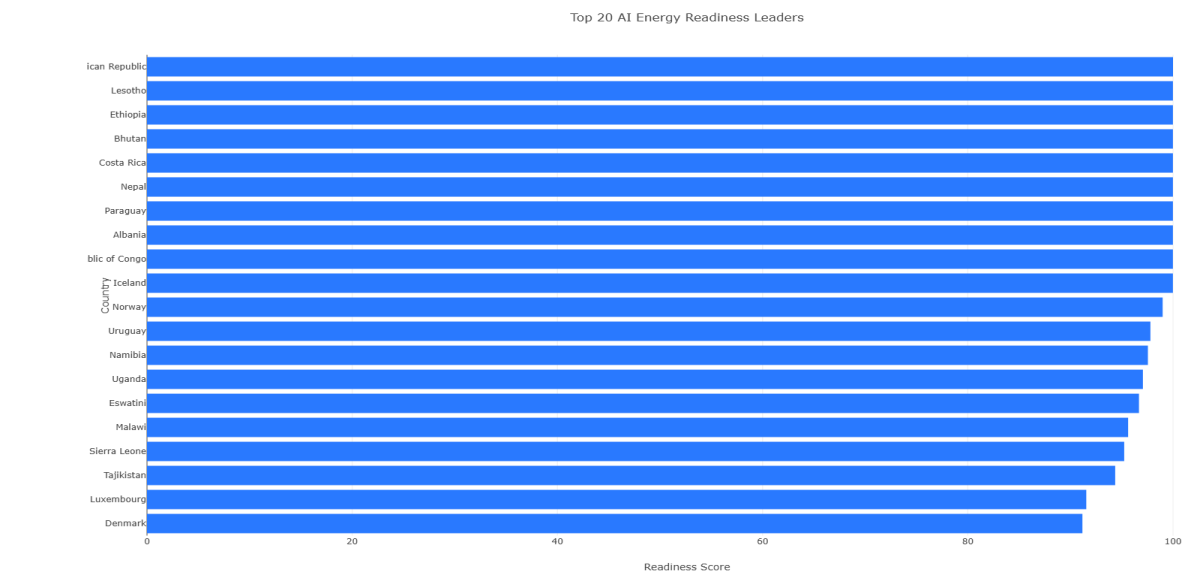
AI Demand Scenarios



AI Scenario Demand Comparison for 2030 / 2035

Scenario	2030	2035
Base	4,353.43	5,567.05
Headwinds	3,137.42	3,379.03
High Efficiency	3,771.30	4,709.83
Lift-Off	5,739.45	7,924.06

AI Energy Readiness & Rankings



Top 20 Readiness Leaders

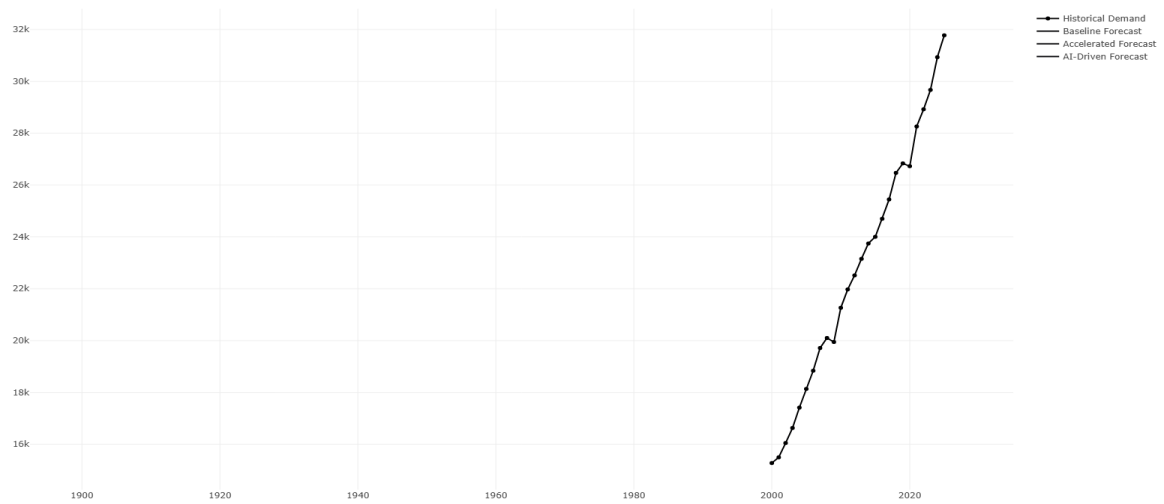
Country	Score	Category
Central African Republic	100.00	Leading
Lesotho	99.99	Leading
Ethiopia	99.99	Leading
Bhutan	99.99	Leading
Costa Rica	99.99	Leading
Nepal	99.99	Leading
Paraguay	99.99	Leading
Albania	99.99	Leading
Democratic Republic of Congo	99.99	Leading
Iceland	99.99	Leading
Norway	98.99	Leading
Uruguay	97.80	Leading
Namibia	97.55	Leading
Uganda	97.06	Leading
Eswatini	96.68	Leading
Malawi	95.63	Leading
Sierra Leone	95.24	Leading
Tajikistan	94.36	Leading
Luxembourg	91.55	Leading
Denmark	91.17	Leading

Grid Stress Index & Regional Risk Assessments

Grid stress reflects fossil dependency, reserve margins, and carbon intensity across countries.

Country	Grid Stress Index	Stress Category
Macao	397.18	High Stress
Palestine	388.48	High Stress
Afghanistan	293.94	High Stress
Djibouti	231.00	High Stress
Turkmenistan	199.71	High Stress
Niger	196.63	High Stress
Togo	194.93	High Stress
Luxembourg	189.74	High Stress
Benin	188.32	High Stress
Kosovo	181.01	High Stress
Montserrat	180.00	High Stress
Falkland Islands	180.00	High Stress
Saint Helena	180.00	High Stress
Botswana	179.47	High Stress
Uzbekistan	173.60	High Stress

Forecast Projections to 2035



Infrastructure & Energy Transition Capacity Requirements

Incremental AI load in 2035 requires approximately 896.9 GW of wind capacity and 1,345.3 GW of solar capacity.

Technology	Capacity Factor	Required Gw
Wind	30%	896.9
Solar	20%	1,345.3

Strategic Recommendations

- Governments should accelerate renewables deployment and grid flexibility programs.
- Utilities must integrate AI load forecasts into resource planning and demand-side management.
- Investors should prioritize low-carbon infrastructure in high readiness, low stress regions.
- AI companies should coordinate with system operators to support managed load growth.

Methodology & Data Sources

Data and analytics use a clean SQLite dataset plus established energy sources. The underlying schema maps clean_energy, clean_ai_demand, and clean_price tables for electricity demand, AI scenario loads, and price trends.

Sources include industry-standard datasets from EIA, IEA, and internal clean energy analytics feeds.

Automated Alert Parameters: Grid Stress Index thresholds, AI demand growth triggers, low carbon share alarms, and demand growth tolerances.